

WIRELESS COMMUNICATION · PREP & STUDY NOTES

5G & the Road to 6G

Slide-by-slide: what's on screen · the technical background to learn · the talk track

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26 slides · 4 parts + open floor · Companion to [index.html](#)

How to use this guide

Each slide has up to three sections. Read the **blue boxes** first to actually understand the material; the talk track is just how to deliver it.

ON SCREEN

What the audience sees — the slide's visible content.

LEARN THE TECH

The real explanation behind the slide — terms, how it works, why. This is the part to study.

TALK TRACK

What to say out loud. Expand naturally; don't read it.

- **The anchor is slide #5 — the five generations.** If you only deeply learn one thing, learn how multiple access evolved (FDMA → TDMA/CDMA → WCDMA → OFDMA) — it explains every later jump.
- **Three acronyms unlock half the deck:** eMBB / URLLC / mMTC (5G's three service classes) and UE / RAN / Core (the three domains). Memorize these.
- **Keep the headline numbers honest:** 20 Gbps, 1 ms, ~1 Tbps are ITU/research *targets*, not what a phone shows.
- **Your edge is the radio/hardware layer** (slides #11–13). Learn those one level deeper than the rest.

01 HERO · TITLE

5G & the Road to 6G

ON SCREEN

- Title + animated signal waveform; tagline “from intelligent networks to intelligence-native networks”; your name & role.

TALK TRACK

Welcome, introduce yourself, set the agenda in one breath: evolution → how 5G works → 6G outlook → careers. Invite questions throughout, open floor at the end.

02 OVERVIEW

What we'll cover

ON SCREEN

- Four areas: evolution of mobile networks · understanding 5G · the road to 6G · careers in telecom & networking.

TALK TRACK

A 20-second signpost. Four areas, then open Q&A.

03 PART ONE

Evolution of Mobile Networks

ON SCREEN

- Opener. Thesis: “Every generation existed to solve the hard limit of the one before it.”

LEARN THE TECH

The honest engine behind every generation is **spectral efficiency** — bits per second squeezed out of each hertz of spectrum (bits/s/Hz). You can grow capacity three ways: more **spectrum** (wider bands), better **efficiency** (smarter modulation/coding), or more **spatial reuse** (smaller cells, more antennas). Each generation pulled all three levers harder than the last.

TALK TRACK

Plant the thread that carries the whole talk and call back to it on every later slide.

04 EVOLUTION · WHY IT MATTERS

Connectivity is infrastructure now

ON SCREEN

- From letters to instant global comms; internet in every pocket; an outage halts business — critical infrastructure, not convenience. Lands in business, healthcare, education, entertainment.

TALK TRACK

Frame connectivity as infrastructure. Tie the sector examples to whatever this room works with.

One signal, five generations

ON SCREEN

- **1G** 1980s analog voice · **2G** 1990s digital + SMS · **3G** 2000s mobile internet · **4G** 2010s LTE broadband · **5G** 2020s connected ecosystem.

LEARN THE TECH — WHAT ACTUALLY CHANGED EACH GENERATION

1G — Analog, **FDMA** (each call gets its own frequency channel). Voice only, circuit-switched (a dedicated line held for the whole call). Systems: AMPS, NMT.

2G — First **digital** generation. **TDMA** (GSM) and **CDMA** (IS-95) let many users share spectrum by time slots or codes. Added SMS; digitized voice = clearer, more secure, more capacity. Packet data bolted on later as GPRS/EDGE (“2.5G”).

3G — **WCDMA / UMTS**. Packet-switched data becomes first-class → real mobile internet. HSPA pushed it to tens of Mbps.

4G — **LTE**, built on **OFDMA** (data split across thousands of narrow orthogonal subcarriers — robust against interference, very efficient). Fully **all-IP**: even voice is data (VoLTE). MIMO introduced. ~100 Mbps–1 Gbps.

5G — **NR (New Radio)**. Keeps OFDM but adds **flexible numerology** (selectable subcarrier spacing) and two frequency ranges: **FR1** sub-6 GHz and **FR2** mmWave (24 GHz+). Massive MIMO + beamforming. Built for three very different traffic types at once (see slide 9).

The one-line story: the access method evolved **FDMA** → **TDMA/CDMA** → **WCDMA** → **OFDMA**, letting each generation pack more independent users and bits into the same spectrum.

TALK TRACK

The deck’s anchor — slow down. Voice → text → internet → video → machines. Point at the denser waveforms: each carries a *new kind* of information, not just more speed.

If short on time: protect this slide.

Why networks never stand still

ON SCREEN

- Growing data consumption · explosion of devices (IoT) · real-time expectations (gaming, AR/VR, control) · automation & M2M.

LEARN THE TECH

These four map onto measurable network demands: **throughput** (data volume), **connection density** (devices/km²), **latency** (real-time response), and **machine-type traffic** (lots of tiny, bursty messages). A single network optimized only for fast phones can’t serve all four — which is exactly the problem 5G’s three service classes solve.

TALK TRACK

These forces compound, so demand never settles. Connected devices already outnumber people.

What the leaps actually feel like

ON SCREEN — DOWNLOAD A 2-HOUR MOVIE (~5 GB)

NETWORK	TIME	THROUGHPUT
3G	2–3 hours	~2–5 Mbps
4G	15–30 minutes	~20–50 Mbps
5G	Seconds to a minute	hundreds of Mbps to Gbps

LEARN THE TECH — THE MATHS SO YOU'RE NOT CAUGHT OUT

Speeds are in **megabits** (Mb), file sizes in **megabytes** (MB). 1 byte = 8 bits, so divide by 8. A 5 GB file = 40,000 megabits. At 50 Mbps: $40000 \div 50 = 800 \text{ s} \approx 13 \text{ min}$. At 1 Gbps: $\approx 40 \text{ s}$. Real downloads are slower than peak (overhead, shared cells, server limits) — which is why the slide gives ranges, not single numbers.

TALK TRACK

A benchmark people feel. Show of hands — who remembers waiting on a 3G download? Bridge: expectations now assume instant, and that pressure drives the next generation.

08 PART TWO

Understanding 5G

ON SCREEN

- Opener. Thesis: “Not a faster 4G — a platform for people, machines, and mission-critical control at once.”

TALK TRACK

Signal you're going under the hood now: what makes 5G different, then where it's used.

What is 5G?

ON SCREEN

- Fifth generation — an intelligent connected ecosystem. Objectives: extreme speed/capacity, ultra-low latency, massive connectivity, mission-critical reliability.

20 Gbps peak (ITU IMT-2020)

~1 ms air-interface latency

10^6 / km² device density

LEARN THE TECH — THE THREE SERVICE CLASSES (THE HEART OF 5G)

eMBB — enhanced Mobile Broadband. Raw speed and capacity: 4K/8K video, AR/VR, crowds in stadiums. *Optimizes for throughput.*

URLLC — Ultra-Reliable Low-Latency Communications. ~1 ms latency and 99.999% reliability: remote surgery, factory robots, vehicle control. *Optimizes for latency + reliability.*

mMTC — massive Machine-Type Communications. Up to ~1 million low-power devices per km²: sensors, meters, trackers. *Optimizes for density + battery life.*

These three pull in opposite directions, so 5G is designed to run all three over **one shared infrastructure** at once — that's the real innovation. The three big numbers on the slide are the ITU **IMT-2020** targets (the official requirements 5G had to meet), not everyday handset figures.

TALK TRACK

Stress that 5G is a *platform*, not a faster phone. Name the three classes plainly. Flag the numbers as ideal targets — saying so builds credibility.

4G versus 5G

CAPABILITY	4G LTE	5G
Peak data rate	~1 Gbps	up to ~20 Gbps
Typical latency	~30–50 ms	~1–10 ms
Device density	~ 10^5 / km ²	~ 10^6 / km ²
Mobility	up to ~350 km/h	up to ~500 km/h
Primary focus	Mobile broadband	Broadband + IoT + critical control

LEARN THE TECH — WHY LATENCY DROPS AND DENSITY RISES

Lower latency comes from: shorter transmission slots via **flexible numerology** (wider subcarrier spacing = shorter symbols), **mini-slots** (send without waiting for a slot boundary), and **edge computing / MEC** (put servers near the user so data travels less distance). **Higher density** comes from massive MIMO serving many users spatially, plus features designed for tiny IoT messages. Note peak speed is the *least* important row for industry — latency and reliability enable control loops.

TALK TRACK

Walk the rows. The game-changers for industry are **latency and density**, not peak speed. Note figures are targets/typical and vary by deployment.

Massive MIMO & beamforming

ON SCREEN

- **Massive MIMO** — many antennas per cell, serves many users at once, more capacity/coverage.
- **Beamforming** — focused, steerable signals aimed per user; less waste, better signal & efficiency.

LEARN THE TECH — YOUR HOME TURF, GO ONE LEVEL DEEPER

MIMO = Multiple-Input, Multiple-Output: antennas at both ends. **Spatial multiplexing** sends *independent data streams* on the same frequency and time, separated only by their spatial path — so N antennas can multiply throughput.

Massive MIMO = a base-station array of 32/64/128+ elements. With **MU-MIMO** it serves many users simultaneously on the same resources by separating them in space.

Beamforming = applying per-element phase/amplitude weights (**precoding**) so the wavefronts add up constructively toward one user and cancel elsewhere — a steerable spotlight vs. a floodlight. Flavours: **analog** (phase shifters, one beam, cheap), **digital** (per-element control, many beams, power-hungry), **hybrid** (the practical mmWave choice).

It all depends on **channel state information (CSI)**. In **TDD**, uplink and downlink share a frequency, so the base station estimates the downlink channel from uplink pilots (**reciprocity**). The resulting **array/beamforming gain** raises SNR — essential to overcome mmWave path loss.

TALK TRACK

MIMO = more antennas, parallel streams. Beamforming = aiming, not broadcasting — floodlight vs. spotlight. This is your world; go off-script with a real example.

An architecture defined by 3GPP

ON SCREEN

- 3GPP standardizes mobile networks. Chain UE → NG-RAN → 5G Core → Data Network, with reference points Uu, N2/N3, N6. Releases: Rel-15 (first 5G, 2018), NR, 5GC, NSA → SA.

LEARN THE TECH — 3GPP, RELEASES, AND NSA VS SA

3GPP (3rd Generation Partnership Project) is a partnership of regional standards bodies that writes the specs every vendor builds to — so a phone works on any operator. Work ships in numbered **Releases**: Rel-15 first 5G NR (2018); Rel-16/17 add URLLC, industrial IoT, RedCap, satellite (NTN); Rel-18+ is branded **5G-Advanced**.

NSA (Non-Standalone): 5G NR radio bolted onto the existing 4G LTE + 4G core — fast to launch, but no full 5G features. **SA (Standalone)**: 5G NR + the new 5G Core — unlocks slicing, URLLC, the works. Most early networks launched NSA and are migrating to SA.

Reference points are just named interfaces: Uu = the air link UE↔gNB; N2 = gNB↔AMF (control signalling); N3 = gNB↔UPF (user data); N6 = UPF↔internet.

TALK TRACK

5G isn't one vendor's design — it's a shared spec in numbered Releases. The arrow labels are the real 3GPP reference points. NSA reused the 4G core; SA is the full 5G core. Walk left to right.

UE, RAN & 5G Core

ON SCREEN

- **UE** — device + 5G modem + (U)SIM. **NG-RAN** — gNB radio, Massive MIMO, split CU/DU/RU. **5GC** — cloud-native, service-based; AMF/SMF/UPF; control/user-plane split.

LEARN THE TECH — WHAT EACH DOMAIN REALLY CONTAINS

UE (User Equipment) — the device, its 5G modem, and the (U)SIM that holds its identity. Cavli's cellular IoT modules are UE. It speaks **NR** over the Uu air interface.

NG-RAN — the radio. The base station (**gNB**) is functionally split: **RU** (Radio Unit: RF + antennas), **DU** (Distributed Unit: real-time lower layers), **CU** (Centralized Unit: higher layers, can be centralized/virtualized). The links are **fronthaul** (RU–DU), **midhaul** (DU–CU), **backhaul** (CU–core). **O-RAN** opens these interfaces so you can mix vendors.

5G Core (5GC) — cloud-native **Service-Based Architecture**: software *network functions* talk via web-style APIs. Key ones: **AMF** (access & mobility), **SMF** (sessions), **UPF** (forwards user data), plus UDM (subscriber data), AUSF (authentication), PCF (policy), NRF (function discovery).

CUPS (control/user-plane separation) lets the UPF sit at the edge near users. Together with SBA, this is what makes **network slicing** possible — multiple logical networks (one tuned for eMBB, one for URLLC, etc.) on one shared physical network.

TALK TRACK

UE is your world. RAN is the radio (gNB, now CU/DU/RU). The Core is cloud-native — AMF/SMF/UPF. The control/user-plane split is what enables slicing and edge breakout.

Where 5G shows up

ON SCREEN

- Smart factories · autonomous vehicles · remote surgery · smart cities · AR/VR · drones, logistics, agriculture.

LEARN THE TECH — MAP EACH TO ITS SERVICE CLASS

Tie examples back to slide 9: AR/VR and 4K streaming = **eMBB**; remote surgery, factory robots, vehicle control = **URLLC**; smart-city sensors, meters, agriculture trackers = **mMTC**. Showing the mapping proves the three classes aren't abstract.

TALK TRACK

Don't read every tile — pick two or three closest to the audience and go deep. Ask what they'd use it for.

What still has to be solved

ON SCREEN

- Challenges: infrastructure cost · spectrum management · security · rural coverage. In production: Private 5G in manufacturing; 5G healthcare.

LEARN THE TECH – SPECTRUM BANDS AND THE COST/COVERAGE TRADE-OFF

5G spans three spectrum tiers, and they trade capacity against reach: **low-band** (<1 GHz) travels far and through walls but carries little; **mid-band** (1–6 GHz, e.g. 3.5 GHz) is the workhorse balance; **mmWave** (24 GHz+) carries enormous data but only over short range and is blocked by walls, rain, even foliage. mmWave therefore needs dense **small cells** — which is the main **cost** driver.

Security: more devices = a bigger attack surface, but 5G also improves it — your permanent identity (**SUPI**) is sent encrypted as a **SUCI**, defeating the IMSI-catcher attacks that plagued older networks, with stronger mutual authentication (5G-AKA).

TALK TRACK

Be honest about cost, spectrum, security, rural. Then balance it: Private 5G is a real, growing segment — a concrete example of where investment flows.

The Road to 6G

ON SCREEN

- Opener. Thesis: “Still research today — but the direction is set: faster, near-instant, intelligence-native.”

TALK TRACK

Shift to looking ahead. Flag clearly that 6G is research today, but the direction is where a lot of current work points.

Why 6G, and what to expect

ON SCREEN

- Why: today’s networks cap out; relentless data growth; immersive AI-driven experiences 5G can’t fully serve.
- Expected: ~1 Tbps (research target), near-zero/sub-ms latency, AI-native, enhanced end-to-end reliability.

LEARN THE TECH – IMT-2030 AND “AI-NATIVE”

Just as 5G = IMT-2020, 6G is being defined as **IMT-2030** by the ITU (vision framework published 2023). **AI-native** is the key phrase: in 5G, AI is bolted on for optimization; in 6G the intent is to *design the air interface and network around AI from the start* — ML doing channel estimation, resource scheduling, and self-management. Treat ~1 Tbps as an aspirational research figure, not a committed spec.

TALK TRACK

Headline: 6G isn’t just faster — it’s **AI-native** with near-zero latency. Frame ~1 Tbps as aspirational to stay credible.

What makes 6G possible

ON SCREEN

- AI in the network · Terahertz comms · Space–air–ground networks · Digital twins.

LEARN THE TECH — THE FOUR ENABLERS, BRIEFLY BUT CORRECTLY

Terahertz (0.1–10 THz) — enormous bandwidth, so huge data rates, but brutal path loss and molecular/atmospheric absorption mean very short range; needs ultra-dense cells and advanced antenna arrays (a hardware frontier).

AI-native networking — ML embedded in radio resource management and the air interface itself (see slide 17).

Space–air–ground / NTN — Non-Terrestrial Networks: LEO satellite constellations, high-altitude platforms (HAPS), and drones integrated with ground networks for true global coverage (5G already started this in Rel-17).

Digital twins — live virtual replicas of a factory or city, fed by real-time network data, to simulate and optimize the physical system. Worth knowing two more 6G buzzwords: **RIS** (programmable surfaces that steer reflected signals) and **ISAC** (the network senses its environment, radar-like, while communicating).

TALK TRACK

Four enablers. THz and space–air–ground are the most physically novel and the hardest hardware problems — your wheelhouse. Digital twins bridge to industry.

The experiences it unlocks

ON SCREEN

- Holographic comms · extended reality · brain–computer interfaces · advanced robotics · autonomous transport · the unimagined.

TALK TRACK

Directional, not promises. Invite the room: which of these would matter in their field?

Research & standardization

ON SCREEN

- Active global research (ITU, 3GPP). 6G being defined now — spectrum, requirements, use cases. ~2030 first commercialization; NOW = standardization.

LEARN THE TECH — THE STANDARDIZATION TIMELINE

Rough roadmap: ITU-R **IMT-2030 vision** done 2023 → detailed technical requirements ~2026–27 → 3GPP **6G study items in Rel-20** (~2025–26) → first **6G specs around Rel-21** (~2028) → commercial launches ~2030. This is the same path 5G took with IMT-2020/Rel-15 — useful to mention so the timeline feels grounded, not hand-wavy.

TALK TRACK

We're in the standardization phase; timelines point to ~2030. Land the opportunity: anyone entering now is early enough to shape it.

Careers in Telecom & Networking

ON SCREEN

- Opener. Thesis: “Everything we covered is built by people — here’s where they sit and how to get there.”

TALK TRACK

The practical part — speak from your own experience.

Why this field

ON SCREEN

- Foundation of everything digital · continuous reinvention (5G/6G/AI/satellites) · durable global demand.

TALK TRACK

Every digital trend — AI and cloud included — rides on this infrastructure, so demand isn't a fad.

Telecommunications roles

ON SCREEN

- RF Engineer · Wireless Comm. Engineer · Core Network Engineer · GNSS & Location · Satellite Comm. Engineer · **Hardware Design** (your role).

LEARN THE TECH — WHAT EACH ROLE ACTUALLY DOES

RF Engineer — antennas, signal chains, RF measurement & testing. **Wireless Comm. Engineer** — LTE/5G protocol stacks, network optimization. **Core Network Engineer** — the 5GC functions and mobility/session management. **GNSS & Location** — positioning and navigation systems. **Satellite Comm.** — NTN / space links. **Hardware Design** (you) — boards, RF front-ends, and the cellular modules that *are* the UE.

TALK TRACK

You sit on the hardware/RF side — offer the day-to-day at a module company like Cavli. The personal angle is what they'll remember.

Networking & emerging roles

ON SCREEN

- Networking: Network · Network Security · Cloud Networking · Data Center Engineer. Emerging: 5G/6G Research · AI for Networks · Private Network · Edge Computing · Network Automation · Cybersecurity.

TALK TRACK

Classic roles plus newer titles from 5G/6G, automation, and AI. Edge computing and cybersecurity are hiring heavily now.

Skills that get you in

ON SCREEN

- Technical: networking fundamentals & TCP/IP · Linux · wireless comms · Python · cloud. Human: problem-solving · communication · collaboration · continuous learning.

LEARN THE TECH — WHY EACH SKILL MATTERS

TCP/IP — everything above the radio is IP; you must know how packets route. **Linux** — base stations, core network functions, and lab tooling run on it. **Python** — test automation, scripting, data analysis of network logs. **Cloud** — the 5G Core is cloud-native (containers, virtualization), so cloud skills now *are* telecom skills. Build hands-on labs (a software-defined radio, a packet capture, a small cloud deployment) — they beat certificates alone.

TALK TRACK

The actionable takeaway. Fundamentals + Linux + Python are the base. Push hands-on projects over certificates.

Let's talk.

ON SCREEN

- Closing. “Questions, perspectives, and where the industry goes next.” Name & role again.

TALK TRACK

Open the floor. Keep seed questions ready. Thank everyone; share contact details if appropriate.

Seed questions: What surprised you most about 5G vs 6G? · Which application matters most in your field? · Hardware vs. software side — which pulls you?

Glossary & acronym cheat-sheet

One-page revision. The starred (★) terms are the ones to know cold.

GENERATIONS & ACCESS METHODS

FDMA	Frequency-Division Multiple Access — each user gets a separate frequency. (1G)
TDMA	Time-Division Multiple Access — users share a frequency in time slots. (2G/GSM)
CDMA	Code-Division Multiple Access — users share spectrum via unique codes. (2G/3G)
OFDMA ★	Orthogonal Frequency-Division Multiple Access — data spread over many narrow orthogonal subcarriers; the basis of 4G & 5G.
GSM / UMTS / LTE	The dominant 2G / 3G / 4G standards, respectively.
NR ★	New Radio — the 5G air interface (the radio standard itself).
VoLTE	Voice over LTE — carrying voice as data packets on 4G (no separate circuit).

5G SERVICE CLASSES & TARGETS

eMBB ★	enhanced Mobile Broadband — high speed/capacity (video, AR/VR).
URLLC ★	Ultra-Reliable Low-Latency Comms — ~1 ms, 99.999% (surgery, robots, vehicles).
mMTC ★	massive Machine-Type Comms — huge density of low-power IoT devices.
IMT-2020	The ITU requirement set that defines “5G” (IMT-2030 = 6G).
ITU / 3GPP ★	ITU = UN body setting the targets; 3GPP = the partnership writing the actual specs (in Releases).

RADIO LAYER

MIMO ★	Multiple-Input Multiple-Output — many antennas sending parallel spatial streams.
Massive MIMO	Base-station array of 32–128+ elements serving many users at once.
MU-MIMO	Multi-User MIMO — separating simultaneous users by their spatial path.
Beamforming ★	Steering signal energy toward a user via per-antenna phase/amplitude weights (a spotlight, not a floodlight).
Precoding	The weights applied across antennas to shape the beam.
CSI	Channel State Information — what the radio knows about the channel; needed to beamform.
TDD / FDD	Time- vs Frequency-Division Duplex — uplink & downlink share time slots vs separate frequencies.
FR1 / FR2	5G frequency ranges: FR1 = sub-6 GHz, FR2 = mmWave (24 GHz+).
mmWave	Millimetre-wave spectrum — huge capacity, short range, easily blocked.

ARCHITECTURE: DEVICE, RADIO, CORE

UE ★	User Equipment — the device + modem + (U)SIM (e.g. a Cavli cellular module).
(NG-)RAN ★	(Next-Generation) Radio Access Network — the radio side.
gNB	The 5G base station (the “g” is for NR/5G; 4G’s was eNB).
CU / DU / RU	Functional split of the gNB: Centralized / Distributed / Radio Unit.
Fronthaul / Midhaul / Backhaul	Links RU–DU / DU–CU / CU–core.
O-RAN	Open RAN — open interfaces so the CU/DU/RU can be mixed across vendors.
5GC ★	5G Core — cloud-native, service-based core network.
SBA	Service-Based Architecture — core functions talk via web-style APIs.
AMF / SMF / UPF ★	Access & Mobility / Session Management / User Plane Function — the three core functions to remember.
UDM / AUSF / PCF / NRF	Subscriber data / Authentication / Policy / Function-discovery.
CUPS	Control/User-Plane Separation — lets the UPF sit at the edge.
NSA / SA ★	Non-Standalone (5G radio on 4G core) vs Standalone (5G radio + 5G core).
Network slicing	Multiple logical networks (each tuned per service class) on one physical network.
MEC	Multi-access Edge Computing — servers near the user to cut latency.
Uu / N2 / N3 / N6	Reference points: air link / gNB–AMF / gNB–UPF / UPF–internet.
SUPI → SUCI	Permanent subscriber identity, sent encrypted as SUCI to stop IMSI-catchers.

6G & BEYOND

IMT-2030	The ITU framework defining 6G (vision published 2023).
AI-native	Designing the network/air interface around machine learning from the start.
THz	Terahertz spectrum (0.1–10 THz) — massive bandwidth, very short range.
NTN	Non-Terrestrial Networks — satellites (LEO), HAPS, drones integrated with ground.
RIS	Reconfigurable Intelligent Surface — programmable surface that steers reflected signals.
ISAC	Integrated Sensing And Communication — the network senses its surroundings while it communicates.
Digital twin	A live virtual replica of a physical system, fed by real-time data.